

Jyothir Anurag Vasu Chinnaboina

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Education

University of Wisconsin–Madison – *B.S. in Computer Science and Data Science*

Madison, WI

Awards: Dean's List (4 semesters)

Expected May 2026

Coursework: Operating Systems, Algorithms, AI/ML, Statistics, Database Management, Data Structures

GPA: 3.7/4.0

Experience

Software Developer Intern (Part-time) – *Vijay Rekha Life Sciences*

Aug 2025 – Present

- Automated 3+ hours of manual work into a single-click data extraction pipeline (Python, pandas, xlswriter) that transforms raw mass spectrometry data for 40+ biomedical compounds into formatted Excel reports.
- Replaced manual reference range lookups with a data processing system (FastAPI, PostgreSQL, pandas) handling 300+ samples daily with automated validation and abnormality flagging across 30+ compounds.
- Accelerated clinical report validation by 85% using a batch processing pipeline (Python, pdfplumber, regex) that extracts and cross-validates patient data from 100+ PDF reports across multiple formats.

Mainframe Intern (Part-time) – *Wisconsin Department of Administration*

Jun 2025 – Present

- Reduced manual triage time by 90% with a Python data pipeline that parsed system logs into structured CSV datasets, flagged remediation status, and routed follow-ups across 5+ teams.
- Eliminated 4+ hours/week of manual reporting via a Python report orchestrator with YAML-driven configuration, JSON Schema validation, Jinja2 templating, and automated email distribution.
- Automated extraction of HTML configuration reports into structured Excel workbooks (Python, BeautifulSoup, xlswriter), processing 44 service classes across 6 policies with forward-fill data normalization.

ERP Integration Intern – *United States Pharmacopeia*

Jun – Aug 2024

- Improved data retrieval time by 70% with an Oracle ERP analytics tool (Python, oracledb, pandas) using optimized SQL queries joining 5 tables to surface order hold trends.
- Automated quarterly stakeholder reports, previously 1 week of manual work, with an ETL pipeline (pandas, Matplotlib, Seaborn) rendering visualizations into Word, PPTX, and PDF for executive distribution.
- Built a web scraping tool (Python, Selenium) that extracted and structured compliance data from internal portals, replacing manual copy-paste workflows across 3 cross-departmental teams.

Projects

Office Hours Queue | *Python, FastAPI, PostgreSQL, Redis, Docker*

Sep 2025 – Present

Co-Founder · Advisor: Prof. Louis Oliphant (CS)

- Reduced student wait times from 1–3 hours to under 15 minutes by launching a data-driven queue platform serving 500+ students with Redis pub/sub messaging and PostgreSQL-backed state management.
- Built a data ingestion pipeline indexing content from 4 sources into a unified ChromaDB vector store with metadata filtering, batch embedding, and per-document fallback on failure.
- Decreased average answer lookup time to ~2s with semantic search (FastAPI, ChromaDB, scikit-learn) across 4 knowledge sources and DBSCAN/KMeans clustering for pattern detection.
- Enabled one-command deployments for 6 services by containerizing with Docker Compose and automating CI/CD via GitHub Actions and Portainer webhooks.

E-Beam 3D Printer Control System | *Python, Node.js, Express*

Jan – Aug 2025

Morgridge Institute for Research (Prof. Kevin Eliceiri)

- Architected a multi-threaded Python data processing system integrating 6+ hardware subsystems over Serial and Modbus RTU with fault-tolerant exponential backoff and graceful degradation.
- Built a real-time data streaming pipeline (Node.js, Express) with log2-based downsampling to aggregate weeks of sensor data across 6 temperature sensors and 13 interlock points for remote analysis.

Supply Chain Forecasting Model | *Python, pandas, scikit-learn*

Jun – Aug 2024

United States Pharmacopeia

- Built a regression-based forecasting model (scikit-learn, pandas) predicting medicine supply chain shortages from 2 years of historical order data, enabling proactive inventory planning across 50+ product lines.

Skills

Languages: Python, Java, SQL, Bash, HTML/CSS

Data & Processing: pandas, NumPy, scikit-learn, BeautifulSoup, Selenium, pdfplumber, ChromaDB, Matplotlib, Seaborn

Frameworks & Tools: Flask, FastAPI, Docker, Docker Compose, Git, GitHub Actions, Pytest

Databases & Cloud: PostgreSQL, Redis, Oracle DB, MySQL, SQLite, Google Cloud Platform

Other: Linux, REST APIs, ETL Pipelines, Data Validation, Web Scraping, Web Crawling, Agile/Scrum, JSON Schema